NE 523: Midterm

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March 11, 2026

1 Integro-Differential Form

1. We begin with the general form for a uniformly-distributed, mono-energetic, fixed source in a pure absorber region.

$$\left[\hat{\Omega} \cdot \nabla + \Sigma_t(\vec{r}, \hat{\Omega}, E)\right] \psi(\vec{r}, \hat{\Omega}, E) = S \tag{1}$$

2. Here, the dependence of Σ_t on $\vec{r}, \hat{\Omega}, E$ can be dropped as $\Sigma_t = \Sigma_a$ is constant over the domain. We can also neglect the dependence of ψ on E because the mono-energetic can only be absorbed – every neutron will either have the source energy or be eliminated.

$$\left[\hat{\Omega} \cdot \nabla + \Sigma_a\right] \psi(\vec{r}, \hat{\Omega}) = S \tag{2}$$

3. For our 2D Cartesian problem, $\hat{\Omega} = \mu\hat{x} + \eta\hat{y}$ Thus, we can express $\hat{\Omega} \cdot \nabla$ as:

$$\hat{\Omega} \cdot \nabla = \mu\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y} \tag{3}$$

4. Using the previous expression, our final "integro"-differential equation becomes:

$$\left[\mu\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y} + \Sigma_a\right] \psi(x, y, \mu, \eta) = S \tag{4}$$

5. As we have vacuum boundary conditions on all sides, we know the incoming flux at each boundary equal 0. Expressing this mathematically gives the following four conditions over the problem region (i.e. $x, y \in [0, a]$):

$$\psi(x = 0, y, \mu > 0, \eta) = 0 \tag{5a}$$

$$\psi(x = a, y, \mu < 0, \eta) = 0 \tag{5b}$$

$$\psi(x, y=0, \mu, \eta > 0) = 0 \tag{5c}$$

$$\psi(x, y=a, \mu, \eta < 0) = 0 \tag{5d}$$

2 PDE to ODE using the Streaming Operator Characteristics

1. By defining $\hat{\Omega} \cdot \nabla$ as the derivative along some characteristic of the streaming operator with a characteristic parameter of s, we can express the streaming term as follows (dependencies of ψ are neglected for brevity):
$$\hat{\Omega} \cdot \nabla \psi = \mu \frac{\partial \psi}{\partial x} + \eta \frac{\partial \psi}{\partial y} = \frac{d\psi}{ds} \tag{6}$$

2. Where x, y can be expressed in terms of the characteristic parameter, s, and the start of the characteristic, (x_0, y_0), as:
$$x = x_0 + \mu s \tag{7a}$$
$$y = y_0 + \eta s \tag{7b}$$
$$\vec{r} = (x, y) = (x_0 + \mu s, y_0 + \eta s) = \vec{r_0} + s\hat{\Omega} \tag{7c}$$

3. We can then rewrite the PDE from part 1 into the desired first-order, ODE.
$$\left[\frac{d}{ds} + \Sigma_a\right] \psi(\vec{r_0} + s\hat{\Omega}, \hat{\Omega}) = S \tag{8}$$

3 Integrating the ODE

1. To integrate the ODE from Part 2 along a general characteristic where $\eta > \mu > 0$, we need to define an integrating factor to multiply by the ODE. The integrating factor is given as:
$$\exp\left(\int_s ds'' \Sigma_a\right) = \exp\left(\Sigma_a \int_s ds''\right) = \exp\left(\Sigma_a s\right) \tag{9}$$

2. Multiply the ODE from Part 2 by the integrating factor (dependencies of ψ are neglected for brevity).
$$\exp\left(\Sigma_a s\right) \frac{d\psi}{ds} + \exp\left(\Sigma_a s\right) \Sigma_a \psi = \exp\left(\Sigma_a s\right) S \tag{10}$$

3. Notice the LHS can be rewritten if we recognize it as a product rule.

$$\exp\left(\Sigma_a s\right)\frac{d\psi}{ds} + \exp\left(\Sigma_a s\right)\Sigma_a\psi = \frac{d}{ds}\left[\exp\left(\Sigma_a s\right)\psi\right] \tag{11}$$

$$\frac{d}{ds}\left[\exp\left(\Sigma_a s\right)\psi\right] = \exp\left(\Sigma_a s\right)S \tag{12}$$

4. Integrate both sides with respect to the characteristic parameter s.

$$\int ds\left\{\frac{d}{ds}\left[\exp\left(\Sigma_a s\right)\psi\right]\right\} = \int ds\exp\left(\Sigma_a s\right)S \tag{13}$$

$$\exp\left(\Sigma_a s\right)\psi = \frac{S}{\Sigma_a}\exp\left(\Sigma_a s\right) + C \tag{14}$$

$$\psi(\vec{r_0} + s\hat{\Omega}, \hat{\Omega}) = \frac{S}{\Sigma_a} + C\exp\left(-\Sigma_a s\right) \tag{15}$$

5. If we say $s = 0$, we are at the start of the characteristic (x_0, y_0), which is some point at the boundary. Evaluating ψ at some boundary point yields:

$$\psi(x_0, y_0, \mu > 0, \eta > \mu > 0) = 0 = \frac{S}{\Sigma_a} + C\exp(0) \tag{16}$$

$$C = -\frac{S}{\Sigma_a} \tag{17}$$

6. This allows us to rewrite ψ for a characteristic such that $\eta > \mu > 0$ as:

$$\psi(\vec{r_0} + s\hat{\Omega}, \hat{\Omega}) = \frac{S}{\Sigma_a}\left[1 - \exp\left(-\Sigma_a s\right)\right] \tag{18}$$

7. For this configuration, we notice the characteristic can spawn (1) on the left side of the region such that $x_0 = 0$ or (2) on the bottom side of the region such that $y_0 = 0$. For both of these cases, we can use the following expression to rewrite the characteristic parameter (s) in terms of either x, y.

$$\frac{x - x_0}{\mu} = s = \frac{y - y_0}{\eta} \tag{19}$$

$$\Rightarrow s = \begin{cases} \frac{y}{\eta} = \frac{x - x_0}{\mu} & \text{if } y_0 = 0 \\ \frac{x}{\mu} = \frac{y - y_0}{\eta} & \text{if } x_0 = 0 \end{cases} \tag{20}$$

8. Using the above expressions for s, we can express ψ in terms of (x, y, μ, η) for a generic characteristic

for which $\eta > \mu > 0$ as given below.

$$\psi(x, y, \mu, \eta) = \frac{S}{\Sigma_a}\left[1 - \exp\left(\Sigma_a \times \begin{cases} \frac{x}{\mu} & \text{if } x_0 = 0 \\ \\ \frac{y}{\eta} & \text{if } y_0 = 0 \end{cases}\right)\right] \tag{21}$$

9. As we will choose some random point in the domain, we need to determine whether x_0 or y_0 will be zero. Using the equations for s, we obtain the following equations:

$$x_0 = 0 \quad \text{if } \frac{y}{\eta} \geq \frac{x}{\mu} \tag{22a}$$

$$y_0 = 0 \quad \text{if } \frac{y}{\eta} < \frac{x}{\mu} \tag{22b}$$

4 Sketching the Solution

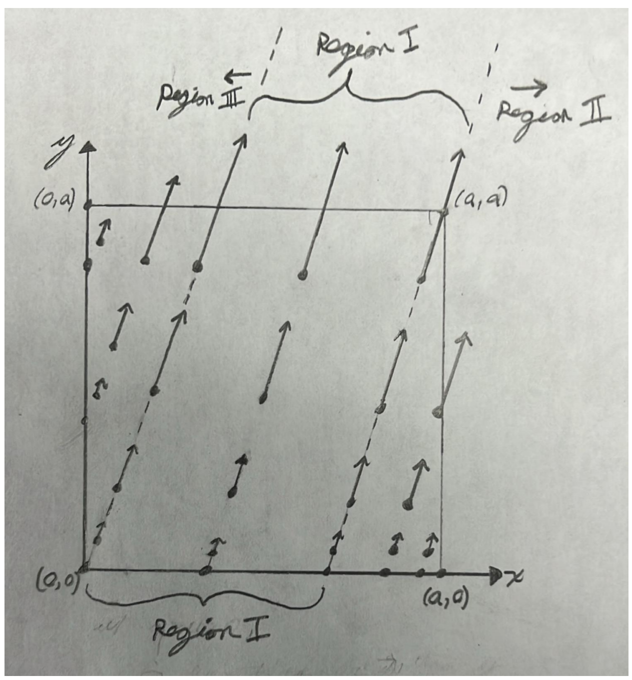

Figure 1: Sketch of the solution over (x,y) plane for an arbitrary $\eta > \mu > 0$

A short description of the sketch:

1. The arrows do not indicate vector quantities. The head of the arrow merely indicates the characteristic direction. The length of the arrow indicates the magnitude of the flux at the given point.

2. As long as the point of interest (the little dots) are inside the The arrows exit the domain freely

3. On the left and bottom sides (these sides are specific to the set of characteristics such that $\eta > \mu > 0$), the magnitude of the flux equals 0.

4. Notice the $\hat{\Omega}$ of the characteristic lies closer to the y-axis than the x-axis. This is set by the condition $\eta > \mu > 0$.

5. In region I, the solution to the flux looks the same for any $x_0 \in$ region I.

6. Regions II and III are symmetric about the line given by the characteristic exactly in the middle x value of region I.

Along the characteristic line $y = x\eta/\mu$, the solution will only have two regions (regions as they are depicted in the sketch). This characteristic line indicates $x_0 = y_0 = 0$. The solution will also be symmetric along the line running from $(0,0) \to (a,a)$. It is worth noting the magnitude of the flux at any given point will only be determined by how far along the characteristic we have traveled (s).

5 Angular Flux by Analogy

1. Before we simplified the expression for s in part 3, we saw the following:

$$\frac{x - x_0}{\mu} = s = \frac{y - y_0}{\eta} \tag{23}$$

2. In these equations, we use (x_0, y_0) to indicate the point where the characteristic ray spawned. For the case $\eta > \mu > 0$, the ray can either spawn (1) on the left side of the region such that $x_0 = 0$ or (2) on the bottom side of the region such that $y_0 = 0$ with the variable not fixed (x if $y_0 = 0$, y if $x_0 = 0$) being allowed to range from $[0, a]$.

3. For case (i) where $\mu < 0, \eta < 0$, the values indicate two spawn locations for the ray (1) on the right side of the region such that $x_0 = a$ or (2) on the top side of the region such that $y_0 = a$. We can then write the flux for case (i) as:

$$\psi(x, y, \mu, \eta) = \frac{S}{\Sigma_a}\left[1 - \exp\left(\Sigma_a \times \begin{cases} \frac{x-a}{\mu} & \text{if } x_0 = a \\ \frac{y-a}{\eta} & \text{if } y_0 = a \end{cases}\right)\right] \tag{24}$$

where

$$x_0 = a \quad \text{if } \frac{y-a}{\eta} \geq \frac{x-a}{\mu} \tag{25a}$$

$$y_0 = a \quad \text{if } \frac{y-a}{\eta} < \frac{x-a}{\mu} \tag{25b}$$

4. For case (ii) where $\mu < 0, \eta > 0$, the values indicate two spawn locations for the ray (1) on the right side of the region such that $x_0 = a$ or (2) on the bottom side of the region such that $y_0 = 0$. We can then write the flux for case (ii) as:

$$\psi(x, y, \mu, \eta) = \frac{S}{\Sigma_a}\left[1 - \exp\left(\Sigma_a \times \begin{cases} \frac{x-a}{\mu} & \text{if } x_0 = a \\ \frac{y}{\eta} & \text{if } y_0 = 0 \end{cases}\right)\right] \tag{26}$$

where

$$x_0 = a \quad \text{if } \frac{y}{\eta} \geq \frac{x-a}{\mu} \tag{27a}$$

$$y_0 = 0 \quad \text{if } \frac{y}{\eta} < \frac{x-a}{\mu} \tag{27b}$$